Sheet chapter 2

1- A simple harmonic oscillator takes 12.0 s to undergo five complete vibrations. Find:
(a) the period of its motion, (b) the frequency in hertz, and (c) the angular frequency in radians per second.

2- An ultrasonic transducer used for medical diagnosis oscillates at 6.7 MHz. How long does each oscillation take, and what is the angular frequency?

3- A 0.60-kg block attached to a spring with force constant 130 N/m is free to move on a frictionless, horizontal surface. The block is released from rest after the spring is stretched 0.13 m. At that instant, find the force on the block and its acceleration.

4- At an outdoor market, a bunch of bananas is set into oscillatory motion with amplitude of 20.0 cm on a spring with a force constant of 16.0 N/m. It is observed that the maximum speed of the bunch of bananas is 40.0 cm/s. What is the weight of the bananas in Newton's?

5- The period of motion of an object spring system is 0.223 s when a 35.4 g object is attached to the spring. What is the force constant of the spring?

6- When a 4.25 kg object is placed on top of a vertical spring, the spring compresses a distance of 2.62 cm. What is the force constant of the spring?

7- A man enters a tall tower, needing to know its height. He notes that a long pendulum extends from the ceiling almost to the floor and that its period is 15.5 s. (a) How tall is the tower? (b) If this pendulum is taken to the Moon, where the free-fall acceleration is 1.67 m/s2, what is the period there?

8- A simple pendulum makes 120 complete oscillations in 3.00 min at a location where g =9.8 m/s^2. Find (a) the period of the pendulum and (b) its length.

9- A "second" pendulum is one that moves through its equilibrium position once each second (The period of the pendulum is 2 s). The length of a second's pendulum is 0.9927 m at Tokyo and 0.9942 m at Cambridge, England. What is the ratio of the free-fall accelerations at these two locations?

10- Find the period and frequency of a simple pendulum 1.000 m long at a location where g =9.8 m/s^2.

11- How long is the simple pendulum which makes 120 oscillations in 3.5 min, if the acceleration of gravity at its location is 9.86 m/s^2?